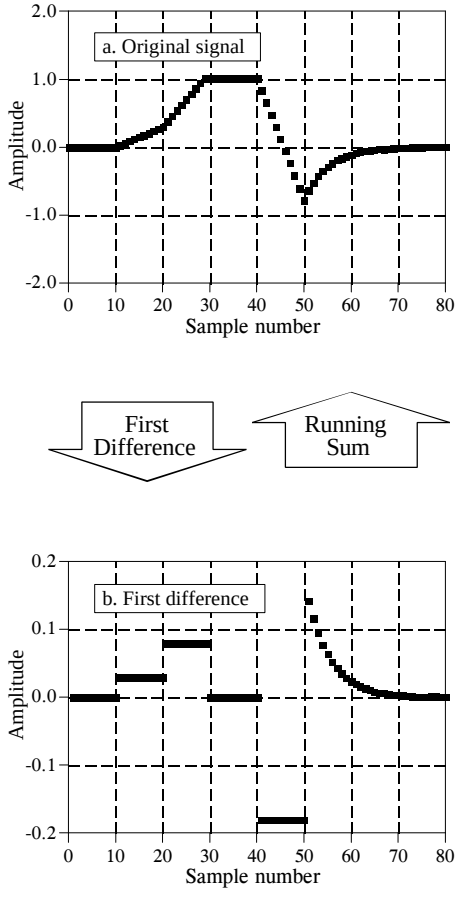

FIGURE 7-3
Example of calculus-like operations. The signal in (b) is the *first difference* of the signal in (a). Correspondingly, the signal is (a) is the *running sum* of the signal in (b). These processing methods are used with discrete signals the same as differentiation and integration are used with continuous signals.

Using this same approach, each sample in the running sum can be calculated by summing all points in the original signal to the *left* of the sample's location. For instance, if $y[n]$ is the running sum of $x[n]$, then sample $y[40]$ is found by adding samples $x[0]$ through $x[40]$. Likewise, sample $y[41]$ is found by adding samples $x[0]$ through $x[41]$. Of course, it would be very inefficient to calculate the running sum in this manner. For example, if $y[40]$ has already been calculated, $y[41]$ can be calculated with only a single addition: $y[41] = x[41] + y[40]$. In equation form:

EQUATION 7-5
Calculation of the running sum. In this relation, $x[n]$ is the original signal, and $y[n]$ is the running sum.

$$y[n] = x[n] + y[n-1]$$

Relations of this type are called **recursion equations** or **difference equations**. We will revisit them in Chapter 19. For now, the important idea to understand is that these relations are *identical* to convolution using the impulse responses of Fig. 7-2. Table 7-1 provides computer programs that implement these calculus-like operations.